Ninad Alurkar

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Education

University of Michigan–Dearborn Sep 2024 – Dec 2025
Master of Science in Robotics Engineering (High Distinction)
GPA: 3.96 / 4.00
Relevant Coursework: Vehicle Mobility Systems (ADAS, autonomy levels, vehicle dynamics, sensing, localization, planning, control), Robotics Perception, Advanced Control Systems

National Institute of Technology–Surat Jun 2018 – Apr 2022
Bachelor of Technology in Mechanical Engineering
CGPA: 8.60 / 10.00

Technical Skills

Programming: C++, Python, MATLAB, Simulink
ADAS / Robotics: ROS1, ROS2, perception pipelines, SLAM, sensor fusion, MoveIt
Simulation & Validation: Gazebo, RViz, MuJoCo (CARLA-transferable workflows), closed-loop testing
Machine Learning: PyTorch, TensorFlow, deep learning, reinforcement learning (PPO), LSTM, Transformers
Perception & Systems: OpenCV, CUDA, Linux, SolidWorks, ANSYS

Research and Projects

Reinforcement Learning–Based Control in MuJoCo Simulation Jan 2026

- Developed a mobile-robot autonomy environment in MuJoCo and trained a PPO controller, improving task success from 28% to 71%.
- Reduced average time-to-goal by 62% (8.7 s to 3.3 s) versus a rule-based baseline, demonstrating performance gains under closed-loop control.

ML-Based Ridge Regression (L2) Strategy for Time-Series Prediction Jan 2026

- Designed a rolling-window Ridge Regression pipeline that reduced out-of-sample error by 15% and improved robustness under covariate shift.

Trust Building in Human–Robot Interaction (Master's Thesis) Jan 2024 – Jan 2026

- Designed and implemented a companion robotic system with adaptive multimodal interaction, building a full ROS architecture on a Clearpath JACKAL integrating an OAK-D Pro camera and ReSpeaker microphone array.
- Integrated vision and audio into a synchronized perception pipeline operating in real-time.
- Conducted an IRB-approved human study with 60 participants, analyzing trust and system reliability using Cronbach's alpha and statistical performance metrics.

Temporal Prediction Models: LSTM vs. Transformer Sep 2025 – Dec 2025

- Built GPU-accelerated forecasting pipelines in PyTorch and TensorFlow, benchmarking models using RMSE and MAE across varying volatility regimes.

Autonomous Manipulation Using Computer Vision Jan 2025 – Mar 2025

- Designed and simulated a closed-loop perception–planning–control system in ROS and Gazebo using OpenCV-based vision, reducing manual task time by 7 minutes.

Professional Experience

Software Engineer Intern Feb 2026 – May 2026
OpenQQuantify, Falls Church, VA (Remote)

- Contributed to development of AI-enabled media platforms and software systems supporting AI-driven electronic design.
- Implemented AI/ML-based components and assisted in development and optimization of AI-integrated electronic systems.

Graduate Research Assistant Jan 2025 – Aug 2025
University of Michigan–Dearborn

- Led an undergraduate team developing a JACKAL UGV for autonomous greenhouse operation, implementing ROS-based LiDAR SLAM and multi-sensor perception.

- Built and validated a computer vision and ML pipeline achieving 98% classification accuracy, evaluated through Gazebo simulation and RViz visualization.

Software Engineer Aug 2022 – Aug 2024

Dover Corporation, Bengaluru, India

- Developed Python- and SQL-based data processing and validation pipelines, reducing manual engineering review effort by 40%.
- Collaborated with mechanical and systems engineers to support sensor-data simulation workflows for system validation.